VIVSHWAN KRISHNA TOMAR

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EDUCATION

Vellore Institute of Technology — Nov 2023 – Present
B.Tech in Computer Science and Engineering with specialization in AIML — *CGPA: 8.20*

The Tribhuvan School — Patna
Higher Secondary Certificate – 81% — *June 2023*

The Tribhuvan School — Patna
Secondary School Certificate – 85% — *June 2021*

TECHNICAL SKILLS

Programming Languages: C++, Python

Libraries & Frameworks: Natural Language Processing (NLP), NumPy, Pandas, Flask, Matplotlib, Tensorflow, Power BI, Django, Scikit-learn

Databases & Tools: MySQL, Git, GitHub, MongoDB Atlas

PROJECTS

WebGuard | ***Python, Pandas, NumPy, Flask, scikit-learn, MongoDB Atlas*** — Nov 2025 – Jan 2026

- Developed a supervised ML classifier to detect malicious vs. legitimate websites using 30+ security and behavioral features including SSL state, domain age, DNS records, web traffic.
- Engineered a scalable ETL pipeline with MongoDB Atlas, successfully processing and validating 10,000+ records for model training and real-time inference.
- Achieved 89% classification accuracy with balanced precision and recall, evaluated using F1-score and confusion matrix.
- Deployed an end-to-end inference system using Flask REST APIs, delivering real-time predictions with sub-200 ms latency.

AgriPredict | ***Python, Flask, TensorFlow, XGBoost, scikit-learn, Matplotlib*** — Feb 2026 – Mar 2026

- Developed a hybrid time-series forecasting system combining LSTM neural networks and XGBoost regressor to predict agricultural crop prices with multiple future time horizons.
- Engineered automated feature extraction pipeline including lag variables (lag_1, lag_2), rolling statistics (mean, standard deviation), temporal features (day of week, month, day of year), and price differentials for robust time-series modeling.
- Implemented comprehensive model persistence system supporting three training configurations (LSTM / XGBoost / Hybrid ensemble) with automatic feature engineering and MinMaxScaler data preprocessing.
- Deployed end-to-end Flask web application featuring interactive dashboard, real-time prediction API, dynamic Matplotlib visualizations, and stakeholder-specific recommendation engine serving farmers, consumers, and government agencies.
- Applied weighted ensemble averaging (50-50) with random noise injection for realistic market simulation and improved forecast accuracy.

CERTIFICATIONS

Microsoft sc 900
Microsoft — *June 2025*

Google IT Support
Google — *Feb 2026*

EXTRA-CURRICULAR

- Technical Competitions: Active competitive programmer (LeetCode 1423, Codeforces 924) and national-level hackathon participant, advancing to Round 2 of the Adobe India Hackathon and ranking in the top 5% at HackVega.
- Co-authored ”AGRI-PREDICT: AI-Driven Framework for Crop Price Forecasting using ML & Time-Series Analysis” published in IJIRT. | Research Paper Link |
- Sports Leadership & Captaincy: Served as Cricket Team Captain and Coordinator, leading the team to a college tournament victory by fostering unity, managing logistics and leveraging strong communication skills to drive on-field coordination and team morale.